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Research Statement

I am a *category theorist*. Trained in basic mathematics, I acquired an only half-sufficient feel for its natural language; I keep training my ear. I am driven by the desire to reveal the fundamental patterns of mathematics, and **implant the electrode** of structural thinking in its multiple pleasure centers.

Uninterested in muscular displays of computational prowess, a category theorist strives not to prove a theorem, but – in the words of Richard Garner – to ‘explain why said theorem is obvious’.

This perspective shapes my attitude towards mathematics: I engaged with many different things, relishing them all, but without fixating on specialisation. The trope of the **competent man** advocated by R. Heinlein is dear to my heart, and I believe it applies to mathematics. I embody the spirit of a taxonomist, intrigued by the interplay of mathematical dialects converging into an intricate universal language. Through this lens, algebra, logic, geometry, topology, analysis, and mathematical physics appear strikingly similar, if not indistinguishable.

I am not a loner, but I have no interest in being recognized as part of a community. This does not imply a lack of interest or concern: quite the contrary. Above all else, I put religious zealotry in the duty of teaching the very few things I understand *clearly* and *faithfully*.

I enjoy watching the same language describe phenomena and theories as different as number theory, topology/logic, fractals, combinatorics...

1. OVERVIEW OF PAST RESEARCH

1.1. Homotopy theory and algebraic topology. I started my research as an algebraic topologist, since it seemed the most profound application of categorical language available at the time. With my advisor **D. Fiorenza**, I showed that in the setting of stable ∞ -categories the theory of *t-structures* introduced by Beilinson, Bernstein and Deligne in [BBD82] as a tool to study sheaves on stratified manifolds is subordinated to a flexible and expressive calculus of *factorisation systems*.

- On one hand, factorisation systems are one of the oldest and most profitable assets of category theory: the first instance of such a structure was produced by Mac Lane in 1948 [ML48], and the notion later found countless applications in abstract algebra, logic, homotopy theory, higher-dimensional category theory.
- On the other hand, the theory of *t-structures* is a fundamental tool in modern algebraic geometry, and it is also a key ingredient in the theory of *mixed Hodge modules* [PS08], also having fundamental links with arithmetic geometry.

Linking these two apparently disconnected worlds has been the content of two papers [FL16, FLM19] co-authored with my advisor and later extended by one of his MSc students, G. Marchetti. A thorough analysis of *t-structures* on stable ∞ -categories constituted my PhD thesis [Lor16].

1.2. Formal category theory. For the subsequent five years, my research was motivated by the desire to understand how categorical language unveils the deep behaviour of the structures that algebraic topology needs.

The central theorem of my PhD thesis was later generalised to the setting of a generic triangulated category in joint work with **S. Virili** [LV20]; this result is best understood through *Grothendieck derivators*, an inherently 2-categorical approach to homotopy theory. From there, I turned to *formal category theory* [SW78, Woo82, Web07, Gra74].

From that moment on, I was lured into what I consider the most elegant approach to category theory, providing deep insights into categorical logic and algebra [Gui82, Web16, Gui75].

This line of research led to [LL23, DLL19, ALL23], where we try to shed some light on fundamental aspects of category theory, e.g. the theory of accessible and presentable categories, generalised adjoint

functor theorems, and the relation between profunctors and the Yoneda lemma. The first paper was published in *Journal of Pure and Applied Algebra*.

1.3. Coend calculus. For me, one of the most useful concepts in category theory is that of a *coend*. During my PhD I wrote a note on the subject that slowly gained popularity among students and eventually became a book [Lor21a] published by Cambridge University Press in 2021.

The book is a collection of results on coends, and it is intended to put together coherently a wealth of material that has so far been confined to research papers or survey reports in a way accessible to anyone with basic knowledge of category theory. It is also a collection of examples of how coends can be used to conduct formal proofs in category theory (using the ‘calculus of coends’ in the title), and how they can be used to express categorical concepts concisely and elegantly.

For a very long time I had the impression that one can turn the act of ‘conducting proofs in category theory using coend calculus’ into a completely formal statement. Together with A. Laretto, the PhD student I am co-advising, we are making this precise [LLV24].

My experience with coend calculus was decisive in writing the joint work [CEG⁺20]. After an extremely long editorial process, the paper was accepted by *Compositionality*, and along the past four years became a go-to reference among users of category-theoretic approaches to modular data accessors in functional programming.

Together with D. Palombi and F. Genovese, we applied coend calculus to the foundations of trading protocols in a trustless environment; in [GLP21b] we describe *escrows* as a certain subclass of the same ‘profunctor optics’ of [CEG⁺20]. The paper was recently accepted by the *Journal of Financial Technology*.

2. OVERVIEW OF PRESENT RESEARCH

2.1. Categorical semantics of Petri nets. My collaboration with Genovese and Palombi did not stop there. Together, we published a series of papers on the categorical foundations of Petri nets.

- In [GLP21c] we define a new flavour of Petri nets where transitions can only fire a limited number of times: they have a lifespan after which they cannot fire again. Published in *ICGT 2021*.
- In [GLP22] we give a categorical semantics for when a Petri net is *bounded*: starting from a marking, no place ever holds more than a predetermined number of tokens along any firing. Published in *ACT 2021*.
- In [GLP21a] we study a particular variety of hierarchical nets, where firing a transition in a parent net must correspond to an execution in a child net: Petri nets become *nested*. Published in *GCM 2021*.

2.2. Categorical automata theory. This fruitful experience taught me that the theory of state machines is an interesting toy for a category theorist: the French school wrote extensively on the subject [Gui78, Bai75, Gui80, Gui74], and some subsequent work followed [BK81, KR90, KKR83], motivating key concepts as that of a quantaloid or of a category enriched over a bicategory.

Guitart and Bainbridge’s style is, to put it mildly, lacking in clarity; the proofs are often omitted, or vanish behind a gesture; it’s Bourbaki’s spirit haunting applied mathematics. I took the challenge to revive this language, taking its fundamental implication (that a piece of formal category theory *coincides* with the mathematical foundation of ‘abstract state machines’, intended *à la* [EKKK, PA70, AM75b, AM75a]) to its natural consequences, trustful that this would have been a fruitful endeavour.

And fruitful it was: this is the project I am most involved in. Together with A. Laretto, the PhD student I am co-advising, and gradually attracting other people, we started a stream of research dealing with the following projects:

- In [BLLL23a] we study how deterministic (=Mealy) automata organise into a bicategory, building on previous work of Katis, Sabadini and Walters; we link their bicategory of ‘processes’ to a bicategory of Mealy machines constructed in 1974 by R. Guitart. We then define Mealy and Moore machines inside a bicategory $\mathbb{B}$, specialising to choices of $\mathbb{B}$ like categories, relations, and profunctors. Accepted for ACT 2023.
- In [BLLL23b] we give a slick proof of completeness and cocompleteness for categories of generalised automata in the sense of [AT90]; a pleasant selling point is certification: we mechanised some main results in the proof assistant Agda. Accepted for CALCO 2023.
- In [BFL⁺23] we study the *semibicategory* of Moore automata: like a bicategory, but lacking identity 1-cells. We show this is far from pathological, and build local adjunctions between this semibicategory and the genuine bicategory of Mealy automata of [BLLL23a].
- In [Lor24] I study Mealy (and, to a minor extent, Moore) monoidal automata where the base monoidal category is that of Joyal’s *combinatorial species*. Species are my favourite category, and the theory of *differential 2-rigs* [LT21] intertwines automata theory at a deep level (which I hope to unveil in the near future).
- In [Lor25a] I study monads in a *double category* of Mealy automata, not (only) for the fun of it, but because one can recognize them as a gadget well-known to representation theorists (especially in the area of Hopf algebra theory).
- In [Lor25b] I push further the idea that double categories are useful to study automata, studying a categorification of transducers [Cho04] in a double category of *two-dimensional transducers*; there are plenty of reasons to study this gadget if you’re a category theorist!

3. PLANS FOR FUTURE RESEARCH

Together with my co-authors, we are writing an unabridged version of [BFL⁺23]. In joint work with D. Castelnovo, we plan to provide a proof of the celebrated Krohn-Rhodes theorem, derived from first principles of 2-category theory and formal enough to extend to a larger class of bicategories of automata.

The experience of B. Femić with higher categories has been decisive in formulating the conjecture that bicategories of Mealy automata enjoy properties akin to a free completion (more specifically, completions under Eilenberg–Moore objects); this is in line with, and a potent generalisation of, prior work by B. Paré [Par10]. We are working on a paper describing this result and its applications.

An exciting open problem from [BLLL23a] is the following. Given a monad T on $\mathbf{Set}$ and a quantale $\mathcal{V}$ [EGHK18, Ch. 2], we can form the locally thin bicategory of $(T, \mathcal{V})$ -profunctors as in [HST14, Ch. III]. As the pair $(T, \mathcal{V})$ varies, one recovers a plethora of bicategories, yielding the categories of topological spaces, approach spaces [Low97], metric and ultrametric spaces, closure spaces... as the $(T, \mathcal{V})$ -categories of [HST14, §III.1.6]. Instantiated in $(T, \mathcal{V})$ -profunctors, the general theory of [BLLL23a, §3] gives a 2-categorical way to look at topological, metric, and (loosely speaking) ‘fuzzy’ approaches to automata theory.

Together with T. Trimble, I co-authored a paper on the theory of *differential 2-rigs* [LT21]. In simple terms, a differential 2-rig is a monoidal category $\mathcal{R}$ with coproducts, such that all functors $A \otimes _ \dashv _ \otimes B$ preserve sums, and equipped with a *differential endofunctor* $\partial : \mathcal{R} \rightarrow \mathcal{R}$ preserving sums and satisfying the Leibniz property $\partial(A \otimes B) \cong \partial A \otimes B + A \otimes \partial B$.

I started studying this concept in complete solitude, and attracted Todd when a preliminary version of my work started circulating. This line of research has immense potential to generalise parts of Joyal’s theory of combinatorial species, and it might provide an unexpected connection between polynomial functors, differential algebra, monoidal categories, and combinatorial differential equations.

We are working on a paper laying foundations for the theory of differential 2-rigs. An open problem in this direction (intersecting categorical automata theory) is the following. Let $(\mathcal{R}, \otimes, \partial)$ be a differential 2-rig. Define the category of ∂ -automata following [AT90], and study its features; for example, [BLLL23b] implies at once that the categories of ∂ -Mealy and ∂ -Moore automata on the category of species are complete and cocomplete (in fact, locally finitely presentable). A particularly enticing class of objects are those that ‘solve’ differential equations in $\mathcal{R}$: call $A \in \mathcal{R}$ a *lax Napier object* if it is a ∂ -coalgebra $a : A \rightarrow \partial A$, and a *pseudo-Napier object* if it is equipped with an isomorphism $A \cong \partial A$ (so, in particular, the terminal ∂ -coalgebra is pseudo-Napier, provided it exists).

4. MISCELLANEOUS PLANS

A few of my plans for the future are:

- With time and patience, I’d like to publish all my old preprints on stable homotopy theory [Lor18, FL15]. I think they contain some ideas worth saving. Algebraic topology is not my field any more (likely, it never has been...), so I treat this as a side project.
- I have a decent experience in the theory of factorisation systems; together with coend calculus, I think it is a piece of basic category theory that deserves, and never had, a dedicated monograph. I have a few ideas on how to write such a book.
- I am part of the *ItaCa* circle, the organization of Italian category theorists; I am interested in promoting category theory in Italy, and I am willing to organise seminars, workshops, and conferences on the subject. I am the main organizative force behind the long-term project of writing a monograph on category theory for Italian students. You should have understood, at this point, that I enjoy writing books more than I enjoy research.
- I am interested in ‘peripheral’ ways to apply category theory; for example, in the foundations of living system theory [Lor21b] or in philosophy. Both communities could use an experienced mathematician, and especially the former line of research widely intersects categorical automata theory: cf. [Bai73, War82, FB96, CLG⁺10, MLS09].

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